	AIR UNIVERSITY
	DEPARTMENT OF COMPUTER ENGINEERING
	EXPERIMENT NO: 8

Lab Tittle: Inter-VLAN

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Objectives: This lab exercise is designed for understanding Inter-VLAN routing..

LAB ASSESSMENT:

Attributes	Excellent (5)	Good (4)	Average (3)	Satisfactory (2)	Unsatisfactory (1)
Ability to Conduct Experiment					
Ability to assimilate the results					
Effective use of lab equipment and follows the lab safety rules					

Total Marks:_____ Obtained Mark: _____

LAB REPORT ASSESSMENT:

Attributes	Excellent (5)	Good (4)	Average (3)	Satisfactory (2)	Unsatisfactory (1)
Data presentation					
Experimental results					
Conclusion					

Total Marks: _____ Obtained Marks: _____



DEPARTMENT OF ELECTRICAL & COMPUTER ENGINEERING

Basic Router Configuration

Lab Report: 8

Tittle: Inter-VLAN

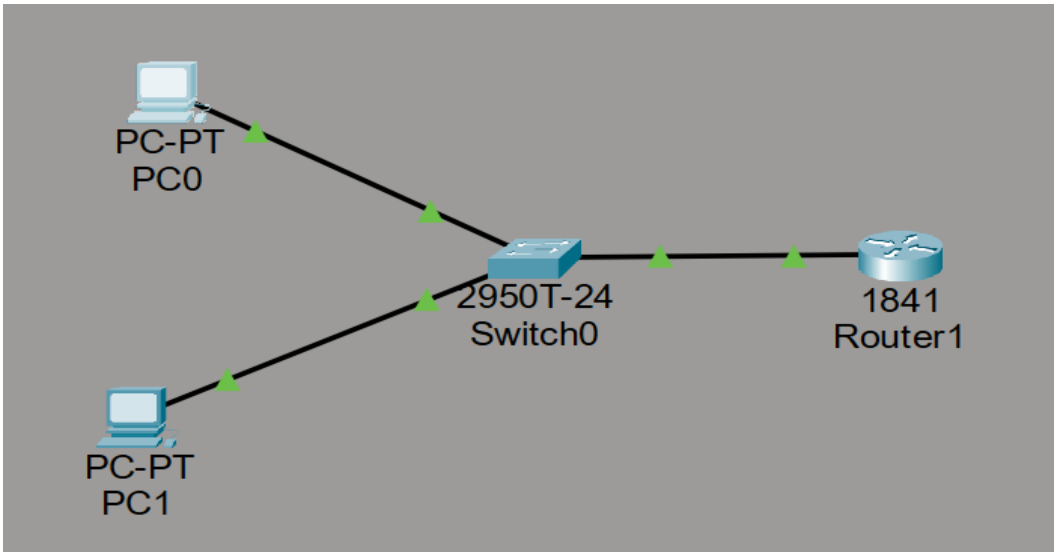
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Submitted To: Sir Basharat Ahmad

Date: 29 Oct, 2026

Task1:



Step 1: Configure Switch Basic Settings

Click **Switch0** > **CLI** tab (or connect via console from a PC) and configure basic passwords and hostname:

Switch Configuration

Physical | Config | CLI | Attributes

```
Switch>
Switch>enable
Switch#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Switch(config)#hostname switch_A
switch_A(config)#enable password 123
switch_A(config)#enable secret 321
switch_A(config)#line console 0
switch_A(config-line)#password 123
switch_A(config-line)#login
switch_A(config-line)#exit
switch_A(config)#
switch_A(config)#line vty 0 4
switch_A(config-line)#password 123
switch_A(config-line)#login
switch_A(config-line)#exit
switch_A(config)#end
switch_A#
%SYS-5-CONFIG_I: Configured from console by console

switch_A#copy running-config startup-config
Destination filename [startup-config]?
Building configuration...
[OK]
switch_A#
```

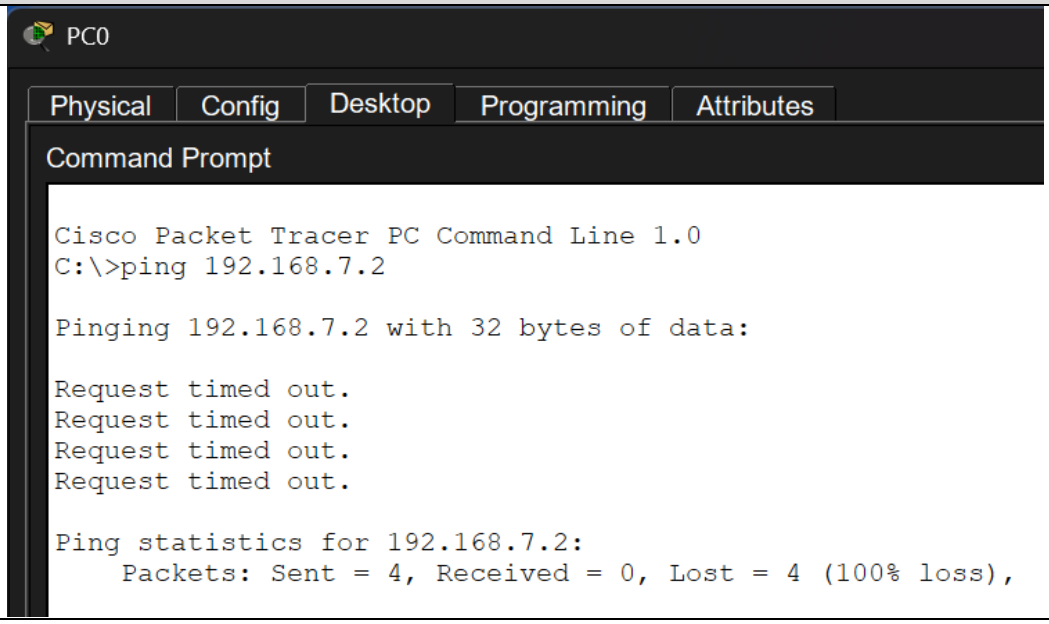
Step 2: Configure Hosts Attached to the Switch

Click each PC > **Desktop** tab > **IP Configuration** and apply the following settings:

Host Device	Switch Port	IP Address	Subnet Mask	Default Gateway	Assigned VLAN
PC0	FastEthernet0/5	192.168.5.2	255.255.255.0	192.168.5.1	VLAN 10 (LAB_A)
PC1	FastEthernet0/9	192.168.7.2	255.255.255.0	192.168.7.1	VLAN 20 (LAB_B)

Step 3: Verify Initial Connectivity

Open Command Prompt on PC0 (192.168.5.2) and run:



Lab Questions & Answers for Step 3:

- **a. Were the pings successful?**
  - **Answer: No.** The pings failed (100% packet loss / Request timed out).
- **b. Why or why not?**
  - **Answer:** PC0 (192.168.5.2/24) and PC1 (192.168.7.2/24) belong to two completely different IP subnets. Because no router interface or default gateway is operational yet to bridge these subnets, Layer 3 traffic cannot pass between them.

Step 4, 5 & 6: Create VLANs & Assign Switch Ports

Configure VLANs and assign access ports on **Switch_A**

```
switch_A#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
switch_A(config)#vlan 10
switch_A(config-vlan)#name LAB_A
switch_A(config-vlan)#vlan 20
switch_A(config-vlan)#name LAB_B
switch_A(config-vlan)#exit
switch_A(config)#
switch_A(config)#interface FastEthernet 0/5
switch_A(config-if)#switchport mode access
switch_A(config-if)#switchport access vlan 10
switch_A(config-if)#exit
switch_A(config)#
switch_A(config)#interface FastEthernet 0/9
switch_A(config-if)#switchport mode access
switch_A(config-if)#switchport access vlan 20
switch_A(config-if)#end
switch_A#
%SYS-5-CONFIG_I: Configured from console by console
```

Step 7 & 8: Display VLAN Interface Information

Run the verification command on Switch_A

```
switch_A#show vlan brief

VLAN Name                             Status    Ports
---- -------------------------------- --------- -------------------------------
1    default                          active    Fa0/1, Fa0/2, Fa0/3, Fa0/4
                                                Fa0/6, Fa0/7, Fa0/8, Fa0/10
                                                Fa0/11, Fa0/12, Fa0/13, Fa0/14
                                                Fa0/15, Fa0/16, Fa0/17, Fa0/18
                                                Fa0/19, Fa0/20, Fa0/21, Fa0/22
                                                Fa0/23, Fa0/24, Gig0/1, Gig0/2
10   LAB_A                            active    Fa0/5
20   LAB_B                            active    Fa0/9
1002 fddi-default                     active
1003 token-ring-default               active
1004 fddinet-default                  active
1005 trnet-default                    active
switch_A#
```

Lab Questions & Answers for Steps 7 & 8:

- **Step 7b: Are there new VLANs in the listing?**
  - **Answer: Yes.** VLAN 10 named LAB_A and VLAN 20 named LAB_B are now listed in the database.

- **Step 8b: Are ports assigned correctly?**
  - **Answer: Yes.** Port FastEthernet0/5 is explicitly bound to VLAN 10 (LAB_A), and port FastEthernet0/9 is bound to VLAN 20 (LAB_B).
  -

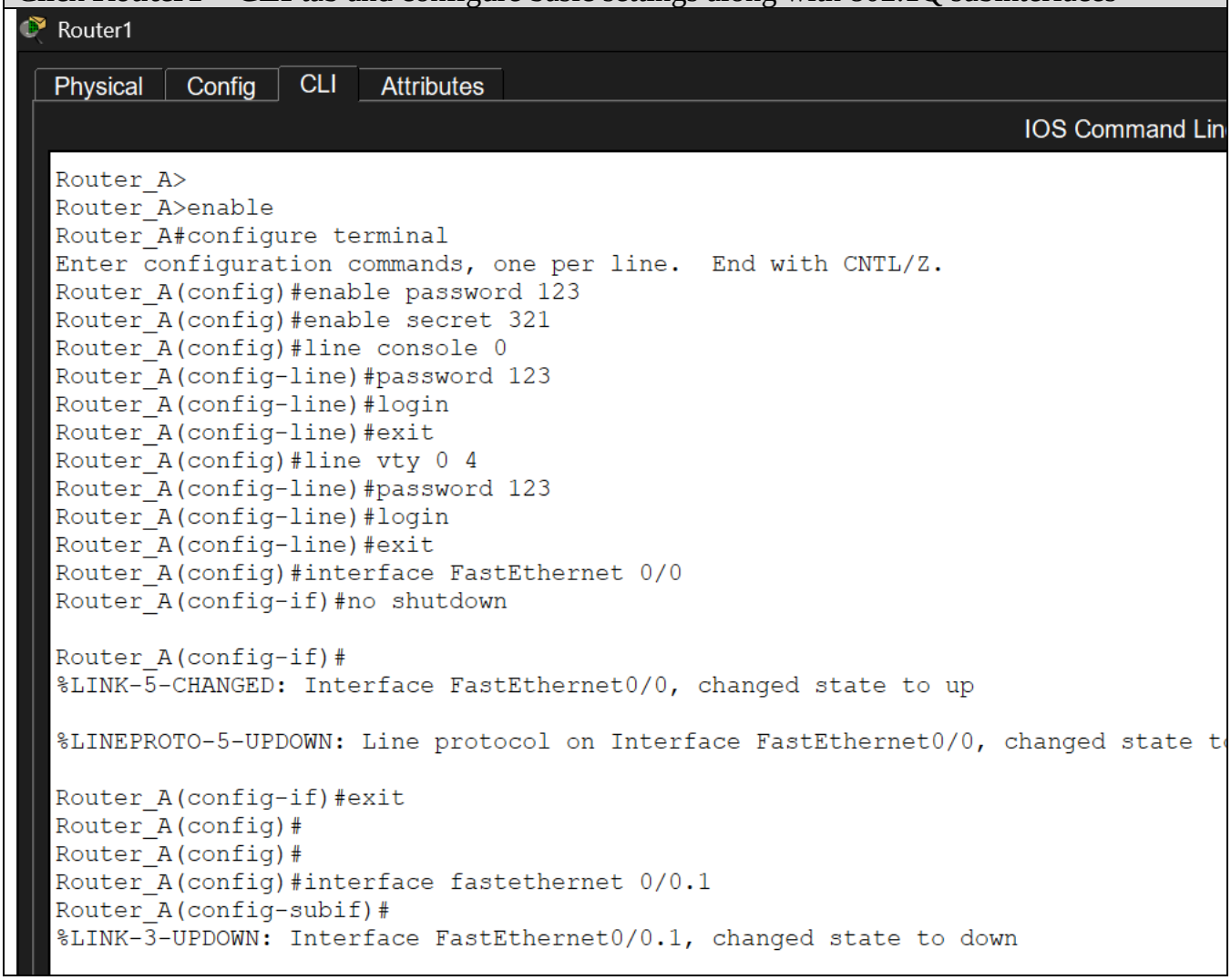
Step 9: Create the Trunk Link on Switch_A

Configure interface **FastEthernet0/1** (connected to Router_A) as an 802.1Q trunk port to carry traffic for both VLAN 10 and VLAN 20

```
switch_A#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
switch_A(config)#interface FastEthernet 0/1
switch_A(config-if)#switchport mode trunk
switch_A(config-if)#end
switch_A#
%SYS-5-CONFIG_I: Configured from console by console
```

Step 10: Configure Router_A (Router-on-a-Stick)

Click **Router1** > **CLI** tab and configure basic settings along with 802.1Q subinterfaces



```
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0.1, changed state to up

Router_A(config-subif)#encapsulation dot1q 10
Router_A(config-subif)#ip address 192.168.5.1 255.255.255.0
Router_A(config-subif)#exit
Router_A(config)#
Router_A(config)#interface fastethernet 0/0.2
Router_A(config-subif)#
%LINK-3-UPDOWN: Interface FastEthernet0/0.2, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0.2, changed state to up

Router_A(config-subif)#encapsulation dot1q 20
Router_A(config-subif)#ip address 192.168.7.1 255.255.255.0
Router_A(config-subif)#end
Router_A#
%SYS-5-CONFIG_I: Configured from console by console

Router_A#copy running-config startup-config
Destination filename [startup-config]?
Building configuration...
[OK]
```

Show ip route

Expected Routing Table Output:

```
C    192.168.5.0/24 is directly connected, FastEthernet0/0.1
C    192.168.7.0/24 is directly connected, FastEthernet0/0.2
```

```
Router_A#show ip route
Codes: C - connected, S - static, I - IGRP, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

C    192.168.5.0/24 is directly connected, FastEthernet0/0.1
C    192.168.7.0/24 is directly connected, FastEthernet0/0.2
```

Step 11: Test Inter-VLAN Routing Connectivity

Open **Command Prompt** on **PC1** (192.168.7.2 on port 0/9) and ping **PC0** (192.168.5.2 on port 0/5):

ping 192.168.5.2

```
Pinging 192.168.5.2 with 32 bytes of data:

Reply from 192.168.5.2: bytes=32 time=5ms TTL=128
Reply from 192.168.5.2: bytes=32 time=6ms TTL=128
Reply from 192.168.5.2: bytes=32 time=6ms TTL=128
Reply from 192.168.5.2: bytes=32 time=6ms TTL=128

Ping statistics for 192.168.5.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 5ms, Maximum = 6ms, Average = 5ms
```

a. Was the ping successful?

- **Answer: Yes.** All ping packets returned successful replies.

b. Why?

- **Answer:** Inter-VLAN routing is fully active. Traffic originating from PC1 (192.168.7.2) travels over the trunk link tagged as **VLAN 20** to **Router_A (Fa0/0.2)**. Router_A performs a Layer 3 routing lookup, removes the VLAN 20 tag, encapsulates the packet with a **VLAN 10** tag, and forwards it back out through **Fa0/0.1** down the trunk link to **Switch_A**, which delivers it to **PC0 (192.168.5.2)**.

Conclusion:

In this lab experiment, I learned how to configure Inter-VLAN routing using a Cisco switch and router in Packet Tracer. At the start, PC0 and PC1 could not ping each other because they were placed in two different VLANs and subnets. To fix this, I created VLAN 10 and VLAN 20 on the switch, assigned the access ports, and set port Fa0/1 as a trunk link. Next, I split the main router interface into sub interfaces and configured 802.1Q encapsulation for both VLAN gateways. I also learned that the trunk port stays down until you enter the no shutdown command on the connected router port. Once all physical interfaces were powered on, the link status changed to up and the ping test between the PCs was successful. Overall, this task made it very easy for me to understand how a router allows devices on different VLANs to talk to each other.